

Solutions of Linear Systems

• The row reduction algorithm leads directly to an explicit description of the solution set of a linear system when the algorithm is applied to the augmented matrix of the system.

• Suppose, for example, that the augmented matrix of a linear system has been changed into the equivalent reduced echelon form.

$$\left[\begin{array}{ccc|c} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \text{The corresponding equations are: (4)}$$

$$1) \ x_1 - 5x_3 = 1 \quad 2) \ x_2 + x_3 = 4 \quad 3) \ 0 = 0$$

The variables x_1 and x_2 corresponding to pivot columns in the matrix are called basic variables. The other variable, x_3 , is called a free variable.

Whenever a system is consistent, as in (4), the solution set can be described explicitly by solving the reduced system of equations for the basic variables in terms of the free variables. This operation is possible because the reduced echelon form places each basic variable in one and only one equation.

$$\begin{cases} x_1 = 1 + 5x_3 \\ x_2 = 4 - x_3 \\ x_3 \text{ is free} \end{cases} \quad (5)$$

The statement " x_3 is free" means that you are free to choose any value for x_3 . Once that is done, the formulas in (5) determine the values for x_1 and x_2 . For instance, when $x_3 = 0$, the solution is $(1, 4, 0)$; when $x_3 = 1$, the solution is $(6, 3, 1)$. Each different choice of x_3 determines a (different) solution of the system, and every solution of the system is determined by a choice of x_3 .

Example 4: Find the general solution of the linear system whose augmented matrix has been reduced to

$$\left[\begin{array}{ccccc|c} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array} \right]$$